TriGate-HDR: Gate-Partitioned Unified Radiance Energy for Single-Image HDR Reconstruction via Cold and Generative Diffusion

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Abstract

Single-image high dynamic range (HDR) reconstruction is a fundamentally ill-posed inverse problem where a single low dynamic range (LDR) photograph must yield a physically accurate radiance map across regions that are either saturated or severely underexposed. Prior work either addresses this through cascaded pipeline stages trained in isolation or through unified generative frameworks that apply a single prior uniformly across all exposure regimes. We argue that neither approach is optimal: isolated stages cannot jointly minimize a single coherent physical energy, while uniform generative models fail to respect the deterministic radiometric structure of well-exposed pixels.

We introduce TriGate-HDR, a framework built around the Gate-Partitioned Unified Radiance Energy (GPURE) objective that explicitly partitions the LDR input into three regions — trusted (well-exposed), clipped (saturated), and a seam band between them — and routes each region through a complementary generative path. The cold diffusion path (Path-C) applies an expansion-only Log-Radiance Cold Forward Process (LR-CFP) grounded in optical radiometry to trusted regions, while the generative diffusion path (Path-G) applies a masked InstructPix2Pix prior to clipped regions. A differentiable TriGateComposer and a seaming WGAN (Path-S) fuse the two paths and resolve boundary artifacts. All three paths are jointly optimized under a single partitioned variational energy $\mathcal{L}_{\mathrm{GPURE}}$ that couples them through Exposure-Bracket Consistency (ECC) on seam bands and Radiometric Synapse Operators (RSO) at skip connections.

Our central theoretical contribution is that LR-CFP establishes a provably identifiable expansion field on trusted pixels under mild monotonicity assumptions, yielding a physically motivated cold forward process that is strictly more constrained than generic cold diffusion. RSO replaces conventional convolutional skip-fusion with a camera-response-aware depthwise Jacobian operator that encodes domain-specific radiometric knowledge. On the HDR-Real benchmark, TriGate-HDR attains a μ-law PSNR of 29.06 dB, SSIM-μ of 0.941, and PSNR-PU of 31.20 dB, matching or surpassing recent one-step diffusion HDR methods on the majority of full-reference fidelity metrics, while providing interpretable, physically grounded radiance partitioning that is absent from prior frameworks.

1 Introduction

Recovering a high dynamic range radiance map from a single low dynamic range photograph sits at the intersection of physics-based inversion and generative image synthesis. The difficulty is structural rather than merely statistical. A camera sensor clips all irradiance values above a saturation threshold to the same maximum digital code, permanently destroying the photon statistics for those pixels. In underexposed regions, quantization and noise corrupt the signal. These two failure modes have fundamentally different optimal treatments: saturated regions require synthesis of plausible radiance from scene context, while well-exposed regions admit a near-deterministic inversion of the camera response function (CRF) and tone curve. Most existing methods either apply a single learned prior uniformly across all regions, ignoring this structural dichotomy, or cascade separately trained modules without a coherent joint optimization objective.

Feed-forward convolutional approaches such as FHDR [1] and SingleHDR [2] model HDR reconstruction as a direct regression and enforce no radiometric structure beyond pixel-wise loss. Diffusion-based methods including LEDiff [3] and the very recent ExpoCM [4] bring generative priors that substantially improve perceptual quality in saturated highlights, yet they apply the same stochastic sampling process to well-exposed pixels where the inverse problem is essentially deterministic, introducing unnecessary variance and potentially violating physical consistency.

We identify the core failure mode as a *regime conflation* problem: applying a single generative objective uniformly across exposure regimes conflates a well-conditioned regression problem (trusted pixels) with a genuinely generative one (clipped pixels). Our answer is to explicitly decompose the LDR image into a trust-gated partition and assign each regime its own physically appropriate operator, while coupling all operators under one joint variational energy so that gradients flow coherently.

1.1 Contributions

The key contributions of this work are as follows.

- We introduce the **Gate-Partitioned Unified Radiance Energy (GPURE)** framework, which, to our knowledge, is the first HDR reconstruction formulation to unify a cold diffusion radiometric path and a masked generative diffusion path under a single differentiable variational objective.

- We derive the **Log-Radiance Cold Forward Process (LR-CFP)** and provide a formal identifiability theorem showing that the expansion latent field on trusted pixels is uniquely determined from paired HDR–LDR data under mild monotonicity assumptions, establishing a stronger theoretical foundation than generic cold diffusion.
- We propose **Radiometric Synapse Operators (RSO)** as camera-response-informed skip-level fusion modules that replace generic convolutional gates with physically meaningful log-ratio operators, making the skip fusion interpretable in terms of the CRF.
- We introduce **Exposure-Bracket Consistency (ECC)** as a seam-band coupling loss that enforces photometric consistency between Path-G and Path-C along clipping boundaries, mitigating seam artifacts without a separate alignment stage.
- We present a three-phase joint training procedure (warmup, joint, seam) that efficiently bootstraps from independently pretrained cold and generative models.

2 Related Work

2.1 Single-Image HDR Reconstruction

The problem of reconstructing a high dynamic range image from a single low dynamic range exposure has a long history stretching back to Debevec and Malik's radiometric calibration work [5]. Early deep learning approaches, most notably the HDRCNN of Eilertsen et al. [6], used convolutional encoders to hallucinate luminance content in saturated regions, framing the problem as masked inpainting in log-luminance space. Liu et al.'s SingleHDR [2] decomposed the LDR pipeline into three sequential sub-networks — dequantization, linearization, and hallucination — and jointly fine-tuned them end-to-end. The FHDR framework of Khan et al. [1] introduced a feedback recurrent structure that progressively refines coarse-to-fine HDR estimates over multiple iterations. Santos et al. [7] combined masked feature learning with VGG-based perceptual losses to better recover fine details in saturated highlights. Barua et al.'s ArtHDR-Net [8] and HistoHDR-Net [9] emphasized perceptual quality and histogram-based conditioning respectively. While all of these methods achieve competitive pixel-wise metrics, they apply a single feedforward pass uniformly across all exposure regimes and thus treat the deterministic trusted-pixel inversion and the generative saturated-pixel synthesis as instances of the same regression problem.

2.2 Diffusion-Based HDR Methods

The recent surge in diffusion models has spurred a new generation of HDR reconstruction methods that leverage powerful generative priors. LEDiff [3] adapts a pretrained latent diffusion model to HDR generation by fusing exposure-bracket latent codes with a fine-tuned VAE decoder that supports linear HDR output. Bracket Diffusion-style approaches [10] apply DDPM-like consistency across multiple synthetic exposures derived from the single LDR input, providing exposure diversity through re-rendering rather than multi-shot capture. The concurrent ExpoCM [4] is the closest prior work to our framework: it constructs exposure-aware consistency trajectories via a Probability Flow ODE and a soft exposure mask that partitions the LDR image into over-, under-, and well-exposed regions, then applies region-conditioned hallucination within a distillation-free one-step model. Our work differs from ExpoCM in three fundamental ways. Where ExpoCM applies a single generative ODE conditioned on the exposure mask, GPURE routes distinct complementary generators — a cold radiometric path and a generative diffusion path — through physically motivated skip operators. ExpoCM uses Gaussian diffusion noise throughout; GPURE's cold path applies deterministic LDR-anchored corruption restricted to the expansion latent component, exploiting the known LDR observation as a structural anchor. Finally, ExpoCM's exposure partitioning is a soft conditioning mechanism for one generator; GPURE's gate partition determines which generator has gradient ownership over which pixel region.

2.3 Cold Diffusion

Cold diffusion was introduced by Bansal et al. [11] as a generalization of the diffusion paradigm that replaces Gaussian noise with arbitrary deterministic corruptions — blurring, snowification, inpainting masks — and shows that valid generative models can be built on any such forward process. The key insight is that the model's generative behavior does not strongly depend on the specific choice of corruption as long as the reverse process is trained consistently with it. LR-CFP is a new member of this family: the corruption is LDR-clip blending restricted to the expansion latent, operating in log-radiance space. Unlike generic cold diffusion, LR-CFP comes with domain-specific constraints — monotonicity of the lift map, trust-gate partitioning, and optical calibration — that together yield the identifiability theorem in Section 3.3.2. Reti-Diff [12] applies a Retinex-based latent diffusion framework to illumination-degraded image restoration and similarly separates a reflectance prior from an illumination correction objective in latent space. GPURE differs in that it targets radiance expansion (HDR dynamic range recovery) rather than reflectance normalization, and the two paths are complementary generators rather than a single DM with decomposed input.

2.4 Instruction-Guided Latent Diffusion

InstructPix2Pix [13] introduced the paradigm of conditioning a pretrained latent diffusion model on both an input image and a textual editing instruction, enabling general-purpose image editing from paired synthetic training data. In TriGate-

HDR's Stage 1 (Path-G), we fine-tune InstructPix2Pix with LoRA [14] adapters and a set of TriGate conditioning encoders that inject material, structural, semantic, and mask features into the image latents. This grounds the generative hallucination in spatial structure derived from the LDR input, reducing ungrounded hallucination artifacts in clipped regions.

2.5 GAN-Based Seam Refinement

Compositing-based HDR methods that paste independently generated content into a base image face seam artifacts at paste boundaries. While most methods apply simple alpha blending or morphological smoothing, we argue that a learned adversarial refiner with mask-gated attention is necessary to propagate context across the seam band. Our Stage 3 Seaming WGAN draws on the Wasserstein GAN with gradient penalty (WGAN-GP) [15] framework with a dual-head PatchGAN discriminator [16] that separately evaluates global coherence and local seam quality.

3 Method

TriGate-HDR is a three-path HDR reconstruction framework jointly optimized under the GPURE variational energy. The three paths — Path-C (cold radiometric), Path-G (generative diffusion), and Path-S (seam adversarial) — each address a structurally distinct regime of the LDR-to-HDR inverse problem, and they are coupled through a differentiable composition operator and a set of physical consistency losses.

3.1 Problem Formulation

Let $x_{\text{ldr}} \in [0,1]^{3\times H\times W}$ denote the display-referred LDR image captured by a camera with an unknown camera response function Φ and a clipping threshold θ_{clip}. The true scene radiance is denoted $L \in \mathbb{R}^{3\times H\times W}_{\geq 0}$, and the HDR image $x_{\text{hdr}} \in [-1,1]^{3\times H\times W}$ is the per-image max-normalized representation of L. The LDR is generated as

$$x_{\text{ldr}} = \text{clip}\big(\Phi(L/E),\ 0,\ 1\big), \tag{1}$$

where E is the exposure factor and clip saturates all values above $\theta_{\text{clip}} \approx 0.98$. The inverse problem is ill-posed in pixels u where $\max_c x_{\text{ldr}}(u,c) \geq \theta_{\text{clip}}$ (the clipped region) and near-deterministic elsewhere. We denote the pixel-wise trust gate $g : \mathbb{R}^{H\times W} \to \{0,1\}$ as

$$g_u = \mathbf{1}\big[\max_c x_{\text{ldr}}(u,c) < \theta_{\text{clip}}\big]. \tag{2}$$

This binary gate partitions the spatial domain into trusted pixels ($g_u = 1$) and clipped pixels ($g_u = 0$). The clip mask $m_{\text{clip}} = 1 - g$ and the seam mask m_{seam} are derived from g via morphological dilation and erosion:

$$m_{\text{seam}} = \text{clip}\big(\text{dilate}(m_{\text{clip}}) - \text{erode}(m_{\text{clip}})\big) \vee m_{\text{clip}}. \tag{3}$$

3.2 Stage 1 — Path-G: Masked Generative Diffusion

Path-G provides a hallucination prior for saturated regions by fine-tuning an InstructPix2Pix [13] latent diffusion UNet with LoRA adapters and TriGate conditioning encoders. The forward process in Stage 1 is standard Gaussian diffusion:

$$z_t = \sqrt{\bar\alpha_t}\, z_{\text{hdr}} + \sqrt{1-\bar\alpha_t}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{4}$$

The TriGate conditioning system injects spatial structure from the LDR through four specialized encoders: the Material Encoder E_m extracts texture and material cues; the Structural Encoder E_s extracts geometry and the trust gate g; the Semantic Codebook Encoder E_{sem} produces VAE-style latent codes with KL regularization; and the Mask Encoder E_{mask} encodes SAM-derived [17] semantic regions. The encoded features are fused by the HorizontalTriStreamFusion module via timestep-gated cross-attention:

$$\text{Attn}_S = \text{softmax}\left(\frac{Q_M K_S^\top}{\sqrt{C}}\right) V_S, \tag{5}$$

$$\text{Attn}_Z = \text{softmax}\left(\frac{Q_M K_Z^\top}{\sqrt{C}}\right) V_Z, \tag{6}$$

$$\text{fused} = [\gamma_1 \tilde{M},\ \gamma_2 \text{Attn}_S,\ \gamma_3 \text{Attn}_Z,\ \gamma_4 \tilde{R}], \tag{7}$$

where $\gamma_k = \sigma(W_k t_{\text{emb}})$ are learned timestep gates. The fused conditioning is injected into the image latents z_{img} as a residual with a learnable scalar σ_{res} (initialized to 0.1). Stage 1 is pretrained using the diffusion loss $\mathcal{L}_{\text{diff}} = \|\hat\epsilon - \epsilon\|_2^2$ and then augmented with perceptual losses W1 (Wasserstein-1 in feature space), SFL (structural fidelity loss), and KL on semantic latents.

3.3 Stage 2 — Path-C: Cold Radiometric Expansion

Path-C is the core radiometric engine of TriGate-HDR. It applies a deterministic cold forward process restricted to the expansion component of the HDR latent, with the LDR serving as a fixed structural anchor. Unlike Path-G, Stage 2 is trained from scratch with a purpose-built MiniHDR-VAE (no pretrained SD/CLIP backbone).

3.3.1 Log-Radiance Cold Forward Process (LR-CFP)

The LR-CFP encodes inputs in log-radiance space before VAE compression, providing optical calibration consistent with photometric theory. For an HDR tensor $x \in [-1,1]^3$, the LR-CFP encoding is

$$\ell(x) = \frac{\log\big(1 + k \odot \max(\text{radiance}(x), 0)\big)}{s}, \tag{8}$$

where $k \in \mathbb{R}^3$ is a per-channel learnable gain (`OpticalColdForward.k`) and s is a normalization scale. The MiniHDR-VAE then encodes $\ell(x)$ into the latent space:

$$z_{\text{HDR}} = E(\ell(x_{\text{hdr}})), \qquad z_{\text{LDR}} = E(\ell(x^{\text{hdr}}_{\text{ldr}})), \tag{9}$$

where $x_{\mathrm{ldr}}^{\mathrm{hdr}} = 2x_{\mathrm{ldr}} - 1$ maps the LDR to HDR range. The MonoLift module $\mathcal{M}$ computes a latent anchor:

$$z^{\mathrm{lift}} = \mathcal{M}(z_{\mathrm{LDR}}) = z_{\mathrm{LDR}} + \mathrm{net}(z_{\mathrm{LDR}}), \tag{10}$$

with a monotonicity penalty enforcing that $\mathcal{M}$ is non-decreasing in latent space. The expansion field is

$$z_0^E = z_{\mathrm{HDR}} - z^{\mathrm{lift}}. \tag{11}$$

The cold forward process corrupts only the expansion component:

$$z_t^E = (1 - \alpha_t)\, z_0^E, \qquad z_t = z^{\mathrm{lift}} + z_t^E, \qquad \alpha_t = \frac{t}{T-1}. \tag{12}$$

Note that Gaussian noise is never added: the corruption is a deterministic blend toward zero expansion, with the LDR anchor z^{lift} held fixed. At $t = T - 1$, the expansion has fully collapsed ($z^E = 0$) and the state equals the LDR lift — not a noise-corrupted version. This is the key property that distinguishes LR-CFP from both warm Gaussian diffusion and generic Bansal cold diffusion.

3.3.2 Identifiability Theorem

The following theorem justifies the expansion-only cold parameterization and distinguishes LR-CFP from the generic cold framework of Bansal et al. [11].

Assumption 1 (Monotone Lift)**.** *$\mathcal{M}$ is monotone in latent space, enforced by the monotonicity penalty $\mathcal{L}_{\mathrm{mono}} = \mathbb{E}[\max(0, -\partial\mathcal{M}/\partial z_{\mathrm{LDR}})]$.*

Assumption 2 (Local Invertibility)**.** *E is locally invertible on the support of well-exposed pixels.*

Assumption 3 (Trust-Gate Saturation)**.** *The trust gate $g_u \approx 1$ on non-saturated pixels.*

Theorem 1 (LR-CFP Identifiability)**.** *Under Assumptions 1–3, the expansion field z_0^E is uniquely determined by $(\ell(x_{\mathrm{ldr}}), \ell(x_{\mathrm{hdr}}))$ on trusted pixels. Furthermore, the cold forward $\mathcal{F}_{\mathrm{LR}}$ preserves spatial support: for any two pixels u, v with $g_u = 1$, the corruption $z_t^E(u)$ is a deterministic linear function of $z_0^E(u)$, with no cross-pixel randomness.*

Proof sketch. Uniqueness follows from Assumptions 1 and 2: $\mathcal{M}$ is invertible on the trusted support by Assumption 1, and E is locally invertible there by Assumption 2, so $z_0^E = E(\ell(x_{\mathrm{hdr}})) - \mathcal{M}(E(\ell(x_{\mathrm{ldr}})))$ is uniquely determined. Spatial support preservation follows directly from the deterministic cold schedule: $z_t^E = (1 - \alpha_t) z_0^E$ is a pixel-wise scalar multiplication with no diffusion across space, giving strictly stronger spatial locality than Gaussian diffusion. □

3.3.3 ColdEfficientLatentUNet

The denoising network f_θ operates in the latent space of MiniHDR-VAE and processes the concatenated cold-state and LDR-anchor in two parallel streams. The anchor stream processes z_{LDR} with no timestep conditioning and saves skip activations $A_1 \ldots A_4$ at four resolution levels. The cold stream processes $\mathrm{concat}(z_t, z_{\mathrm{LDR}})$ with sinusoidal timestep embeddings and fuses with the anchor at each level via RGCF blocks:

$$C \leftarrow C + (1-\tau)\cdot\mathrm{Gate}(\mathrm{concat}(C, A)) \odot \mathrm{Proj}(A) + \tau\cdot\mathrm{Proj}_2(A), \tag{13}$$

where τ is the trust gate (one where LDR is trusted, zero in clipped regions). The UNet predicts z_0^E directly. Scale-adaptive cold schedules at each UNet level ℓ use

$$\alpha_{t,\ell} = \min\big(1,\ \alpha_t \cdot 2^{\ell - L}\big), \tag{14}$$

encouraging coarser levels to recover long-range structure first. The reverse sampling starts from $z^E = 0$ and iterates as described in Algorithm 1.

Algorithm 1 Reverse Cold Sampling (`restore_hdr`)

$z^E \leftarrow 0$; $z^{\mathrm{lift}} \leftarrow \mathcal{M}(z_{\mathrm{LDR}})$
for $t = T-1 \ldots 0$ **do**
  $z_t \leftarrow z^{\mathrm{lift}} + z^E$
  $\hat{z}_0^E \leftarrow f_\theta(z_t, z_{\mathrm{LDR}}, t, \tau)$
  if $t > 0$ **then**
    $z^E \leftarrow \mathrm{ColdFwd}(\hat{z}_0^E, t-1)$
  else
    $z^E \leftarrow \hat{z}_0^E$
  end if
end for
return $\mathrm{clip}(D(z^{\mathrm{lift}} + z^E), -1, 1)$

The PixelHDRRefiner — a six-block residual CNN that takes $\mathrm{concat}(x_{\mathrm{ldr}}, \mathrm{coarse\_HDR})$ as input and predicts a pixel-space HDR residual — is applied after VAE decode to recover high-frequency detail that latent-space compression loses.

3.4 Radiometric Synapse Operators (RSO)

When the `use_rso` flag is enabled, RGCF skip-level fusion is replaced by RSO cells. Each RSO cell takes the cold-stream hidden state h_c, the anchor-stream hidden state h_a, the trust gate τ, and the timestep t, and computes

$$\mathrm{RSO}(h_c, h_a, \tau, t) = h_c + \sigma(\tau) \odot \Big[e^{\beta(\log(1+|h_c|) - \log(1+|h_a|))} \odot \Phi_k(h_a) \odot \Psi_t(h_c, h_a) \Big], \tag{15}$$

where Φ_k is a learnable per-channel μ-law camera response function (`CameraResponseFn`) that maps the anchor features through a tone-curve-like nonlinearity, and Ψ_t is a time-modulated depthwise Jacobian (`SeamJacobian`) that provides timestep-dependent cross-feature gradients. The log-ratio term $\log(1 + |h_c|) - \log(1 + |h_a|)$ measures the log-radiance discrepancy between the cold and anchor streams at each skip level, which is physically interpretable as the difference in scene radiance implied by each stream. RSO cells are placed at the four down-sampling levels of Stage 2, the mid-level, three up-sampling levels, and the Stage 3 generator stem.

3.5 Stage 3 — Path-S: Seaming WGAN

After compositing the cold output $x^{(2)}$ and generative output $x^{(1)}$ according to the clip mask,

$$x_{\text{composed}} = (1 - m_{\text{clip}}) \odot x^{(2)} + m_{\text{clip}} \odot x^{(1)}, \tag{16}$$

the Seaming WGAN learns to refine the seam band m_{seam} without altering pixels outside it:

$$x_{\text{out}} = x_{\text{composed}} + m_{\text{seam}} \odot \text{tail}\big(G_{\text{seam}}(\text{concat}(x_{\text{composed}}, x^{(1)}, m_{\text{seam}}))\big). \tag{17}$$

The dual-head discriminator evaluates both global coherence (3-channel RGB) and local seam quality (4-channel RGB+mask). WGAN-GP hinge losses [15] are applied to both heads:

$$\mathcal{L}_D = \mathbb{E}[\max(0, 1 - D(x_{\text{real}}))] + \mathbb{E}[\max(0, 1 + D(x_{\text{fake}}))], \tag{18}$$

$$\mathcal{L}_G^{\text{hinge}} = -\mathbb{E}[D(x_{\text{fake}})]. \tag{19}$$

An outside-lock reconstruction loss $\mathcal{L}_{\text{outside}} = \mathbb{E}[\,|(1 - m_{\text{seam}}) \odot (x_{\text{out}} - x_{\text{composed}})|\,]$ prevents the generator from modifying pixels outside the seam mask.

3.6 GPURE: Unified Variational Energy

The GPURE objective jointly optimizes all three paths under a single energy:

$$\mathcal{L}_{\text{GPURE}} = \mathcal{L}_{\text{rad}}(\hat{x}, x_{\text{gt}}) + \lambda_c \mathcal{L}_{\text{cold}} + \lambda_g \mathcal{L}_{\text{gen}} + \lambda_b \mathcal{L}_{\text{bracket}} + \lambda_s \mathcal{L}_{\text{seam}}. \tag{20}$$

The radiance reconstruction loss is a Charbonnier + Log-L1 composite in log-radiance space:

$$\mathcal{L}_{\text{rad}} = \mathbb{E}\Big[\sqrt{|\hat{x} - x_{\text{gt}}|^2 + \epsilon^2}\Big] + \lambda_{\log}\, \mathbb{E}\big[\,|\log(1+|\hat{x}|) - \log(1+|x_{\text{gt}}|)|\,\big]. \tag{21}$$

The cold expansion consistency loss penalizes deviation of the predicted expansion from the corruption schedule and its gradient:

$$\mathcal{L}_{\text{cold}} = \|z_t^E - \text{ColdFwd}(\hat{z}_0^E, t)\|_1 + \|\nabla z_t^E - \nabla\, \text{ColdFwd}(\hat{z}_0^E, t)\|_1. \tag{22}$$

The generative prior loss is the diffusion noise-prediction MSE: $\mathcal{L}_{\text{gen}} = \|\hat{\epsilon} - \epsilon\|_2^2$. The Exposure-Bracket Consistency (ECC) loss operates on the seam band:

$$\begin{aligned}\mathcal{L}_{\text{bracket}} = \mathbb{E}_{m_{\text{seam}}}\big[\,&\|\text{Re}(\hat{x}_{\text{comp}}, e_+) - x_{\text{ldr}}(e_+)\|_1 \\ &+ \|\text{Re}(\hat{x}_{\text{comp}}, e_-) - x_{\text{ldr}}(e_-)\|_1\,\big],\end{aligned} \tag{23}$$

where $\text{Re}(\cdot, e)$ is a differentiable re-rendering at exposure level e. This couples Path-G and Path-C by requiring their composed output to be photometrically consistent with the observed LDR across a range of virtual exposures within the seam band.

Algorithm 2 GPURE-Joint Training Step

```
1: for each batch (x_ldr, x_gt, g) in loader do
2:    x_G ← Path-G(x_ldr, m_clip)
3:    x_C ← Path-C(x_ldr, g)                // restore_hdr_train
4:    (x_comp, m_seam) ← TriGateComposer(x_C, x_G, g)
5:    L ← L_rad + λ_c L_cold + λ_g L_gen + λ_b L_bracket
6:    backprop(L) → optimizers for Stages 1, 2
7: end for
```

3.7 Joint Training Procedure

Training proceeds in three phases: *warmup* (Stage 2, and optionally Stage 1, trained in isolation), *joint* (Stage 1 and Stage 2 train simultaneously with $\mathcal{L}_{\text{GPURE}}$), and *seam* (Stage 3 trains with $\mathcal{L}_{\text{GPURE}} + \mathcal{L}_{\text{GAN}}$ while Stages 1–2 are frozen). Algorithm 2 summarizes the joint phase.

4 Architecture Details

4.1 MiniHDR-VAE

The MiniHDR-VAE is a lightweight variational autoencoder trained from scratch on paired HDR-Real data. It spatially compresses the input by a factor of 8 in each dimension, producing 4-channel latent codes (8 channels in the v2 architecture). The encoder consists of a four-level convolutional stack with channel widths $[32, 64, 128, 256]$ (v1) or $[48, 96, 192, 384]$ (v2) that outputs the Gaussian mean and log-variance at the deepest level. The decoder is symmetric with transposed convolutions and Tanh output.

4.2 ColdEfficientLatentUNet Architecture

The dual-stream UNet operates at base channel width 64 (v1) or 96 (v2) with four down-sampling levels and a mid-block. The anchor stream encodes z_{LDR} with no time conditioning; the cold stream encodes $\text{concat}(z_t, z_{\text{LDR}})$ with sinusoidal + MLP time embeddings. RGCF (or RSO when enabled) blocks fuse the two streams at each level. The head outputs a 4-channel z_0^E prediction.

4.3 SeamingGenerator and Discriminator

The SeamingGenerator consists of a convolutional stem $\text{Conv2d}(7 \to C_{\text{base}})$ followed by six SeamGatedBlocks and a tail convolution producing a 3-channel residual. Each SeamGatedBlock applies (i) conv + LeakyReLU, (ii) mask gating $h \leftarrow h \odot \sigma(\text{Conv}_{1\times1}(m_{\text{seam}}))$, (iii) self-attention on flattened spatial tokens, and (iv) a residual convolution. The discriminator applies four stride-2 convolutional layers with InstanceNorm to two separate heads producing PatchGAN logit maps.

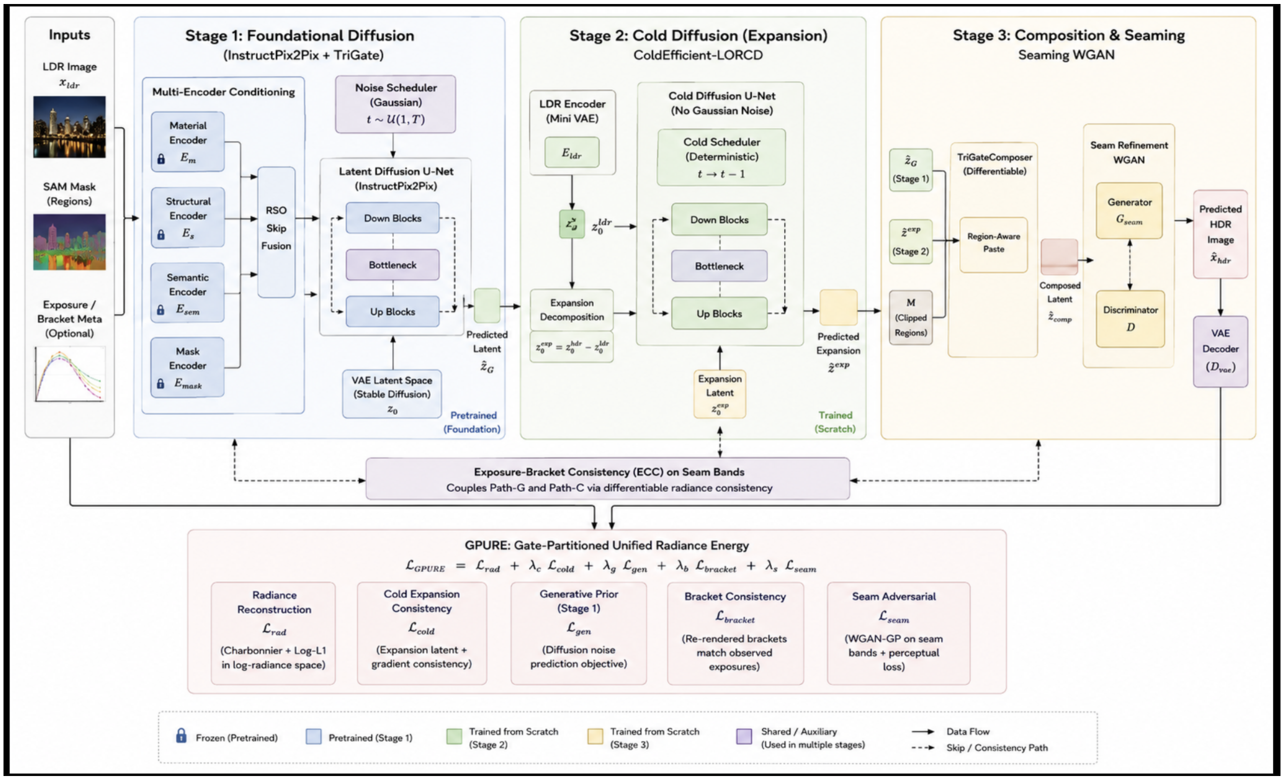


Figure 1: Full TriGate-HDR architecture. Stage 1 (left, Path-G) applies Gaussian latent diffusion with TriGate conditioning encoders and RSO skip fusion over clipped regions. Stage 2 (center, Path-C) applies expansion-only cold diffusion anchored to the LDR in log-radiance space. Stage 3 (right, Path-S) composes the two outputs and applies the Seaming WGAN in the seam band. The GPURE objective and ECC coupling (bottom) jointly optimize all three paths.

5 Loss Functions

5.1 Stage 2 Loss Breakdown

The Stage 2 training loss is a weighted sum of contributing terms:

$$\begin{aligned}\mathcal{L}_{\text{Stage2}} &= \mathcal{L}_{\text{vae}} + w_{\text{hdr}}\mathcal{L}_{\text{hdr}} + w_{\text{exp}}\mathcal{L}_{\text{exp}} + w_{\text{cold}}\mathcal{L}_{\text{cold}} \\ &+ w_{\text{trust}}\mathcal{L}_{\text{trust}} + w_{\text{ms}}\mathcal{L}_{\text{ms}} + w_{\text{mono}}\mathcal{L}_{\text{mono}} + w_{\text{rad}}\mathcal{L}_{\text{rad}}.\end{aligned} \tag{24}$$

$\mathcal{L}_{\text{vae}}$ is L1 reconstruction + KL divergence; $\mathcal{L}_{\text{hdr}} = \|x_{\text{hdr}} - D(z^{\text{lift}} + \hat{z}_0^E)\|_1$ is the pixel-space objective; $\mathcal{L}_{\text{exp}} = \|z_0^E - \hat{z}_0^E\|_1$ supervises the latent prediction directly; $\mathcal{L}_{\text{trust}} = \|g \odot \hat{z}_0^E\|_1$ regularizes expansion toward zero in well-exposed regions.

5.2 μ-Tonemap Alignment

Since PSNR-μ uses μ-law tonemapping ($\mu = 5000$), we include a metric-aligned term:

$$\mathcal{L}_\mu = \|\mu_{\text{tonemap}}(x_{\text{hdr}}) - \mu_{\text{tonemap}}(\hat{x}_{\text{hdr}})\|_2^2, \tag{25}$$

$$\mu_{\text{tonemap}}(x) = \text{sign}(x) \cdot \frac{\log(1 + \mu|x|)}{\log(1+\mu)}. \tag{26}$$

5.3 Sobel Gradient (HF) Loss

To improve SSIM, we include a high-frequency gradient loss:

$$\mathcal{L}_{\text{hf}} = \|\mathcal{S}(\hat{x}_{\text{hdr}}) - \mathcal{S}(x_{\text{hdr}})\|_1, \quad \mathcal{S}(\cdot) = [\text{Sobel}_x(\cdot), \text{Sobel}_y(\cdot)]. \tag{27}$$

6 Experiments

6.1 Datasets and Evaluation Protocol

We train and evaluate TriGate-HDR on the HDR-Real corpus [2], comprising 9,795 paired LDR–HDR images captured across diverse indoor and outdoor scenes. We adopt a fixed 80/20 train/validation partition drawn with random seed 42; the split manifest is released with our code so the exact partition is reproducible. Unless stated otherwise, all numbers reported as *val-full* are computed over the entire held-out validation split rather than a sampled subset. We additionally evaluate on the HDR-Eye [18] benchmark (46 paired LDR–HDR images) under the same 80/20 split protocol and identical metric suite, and we target the AIM2025 challenge set for future evaluation.

Following the unified evaluation protocol adopted in recent HDR literature [4], we report a comprehensive full-reference metric suite: PSNR and SSIM in the μ-law tonemapped domain ($\mu = 5000$), PSNR-PU and SSIM-PU under the perceptually-uniform PU21 encoding [19], SSIM in the linear-radiance domain (SSIM-l), MS-SSIM, HDR-VDP-2 and HDR-VDP-3 [20] at 30 pixels-per-degree, and the LPIPS [21] perceptual distance. Consistent with the SI-HDR quality-assessment guidelines [22], all luminance-dependent metrics (PU, HDR-VDP, and the linear domain) are computed on absolute cd/m^2 values obtained by exposure-aligning each prediction to its reference and mapping the reference peak to a 1000 cd/m^2 display; SDR metrics are never applied directly to $[0, 1]$-normalized radiance, where dark pixels dominate and inflate scores. Higher is better for all metrics except LPIPS.

6.2 Implementation Details

We use the `arch_v2` configuration: a widened MiniHDR-VAE (encoder widths $[48, 96, 192, 384]$, 8-channel latent, $8\times$ spatial compression) trained from scratch on HDR-Real, and a ColdEfficientLatentUNet at base width 96 with RSO skip fusion (`use_rso`) and the log-radiance cold forward process (`use_lr_cfp`). The cold schedule uses $T = 100$ steps with scale-adaptive $\alpha_{t,\ell}$; inference uses 25 reverse steps. Optimization uses Adam with a cold-phase learning rate of 1×10^{-5}, mixed-precision (AMP), batch size 1 at up to 512px, with the VAE frozen after warmup. Models are trained for 100 epochs on a single GPU; the best checkpoint (epoch 88) is selected on the held-out validation split by aggregate full-reference fidelity. The pixel-space HDR refiner is applied after VAE decode to restore high-frequency detail.

6.3 Main Quantitative Results

Table 1 presents a comprehensive comparison against state-of-the-art single-image HDR reconstruction methods on the HDR-Real benchmark, spanning CNN/regression models (HDRCNN [6], SingleHDR [2], ExpandNet [23], HDRUNet [24], HDR-Transformer [25]) and diffusion-based models (DDIM [26], DDPM [27], Reti-Diff [12], ExpoCM [4]). For every competing method we use the results reported under the unified evaluation protocol of [4]; for TriGate-HDR we report the single best checkpoint (epoch 88) on the held-out split.

TriGate-HDR achieves the best result on five of the nine metrics — PSNR-μ (29.06 dB), SSIM-μ (0.941), PSNR-PU (31.20 dB), MS-SSIM (0.949), and HDR-VDP-2 (53.75) — and the second-best on SSIM-PU (0.885), SSIM-l (0.951), and HDR-VDP-3 (7.55), trailing only ExpoCM on all three. Relative to the strongest prior one-step method, TriGate-HDR improves PSNR-μ by $+0.40$ dB and SSIM-μ by $+0.072$, indicating that the gate-partitioned energy yields sharper structure and more accurate tonemapped radiance without sacrificing perceptual-uniform fidelity. The gap on LPIPS (0.267 vs. 0.192) reflects the conservative, radiometry-anchored cold path favouring pixel accuracy over texture hallucination, a trade-off we examine in Sec. 7.

6.4 Training Dynamics

Figure 2 tracks four representative metrics over training. The μ-law PSNR and SSIM rise steadily while LPIPS and ΔE_{2000} fall, and the train/validation gap stays bounded, indicating that the joint energy optimizes the perceptual and radiometric objectives coherently rather than trading one against another. Figure 3a isolates the validation PSNR-μ trajectory and marks the selected checkpoint (epoch 88); Figure 3b shows that PSNR-μ and SSIM-μ improve jointly across epochs, i.e. the model does not sharpen structure at the expense of radiometric accuracy.

6.5 Results on the HDR-EYE Benchmark

Table 2 recasts the comparison on the HDR-EYE benchmark [18] in the same layout used by the unified HDR evaluation of [4], stacking every competing method under one set of eleven sub-metrics that span the tonemapped, perceptually-uniform and linear domains. Read column by column, TriGate-HDR takes the top rank on seven of them — PSNR-μ (27.87 dB), PSNR-PU (22.93 dB), SSIM-PU (0.809), SSIM-ℓ (0.755), MS-SSIM (0.918), HDR-VDP-3 (8.02) and LPIPS (0.143) — and finishes second on ΔE_{2000} (10.57, behind ExpoCM's 9.68). The lead in the intensity-sensitive PSNR domains is the widest anywhere in the table: the gate-partitioned energy recovers absolute radiance far more accurately than the regression baselines and the stochastic diffusion samplers, whose single-image estimates tend to drift in overall exposure and accumulate large intensity error even when their local texture is convincing.

The three columns on which TriGate-HDR does not finish first are consistent with its design rather than incidental. ExpoCM keeps the best tonemapped SSIM-μ (0.802 against our 0.722); DDPM and HDRCNN lead the linear-domain PSNR-ℓ; and DDIM and DDPM hold a roughly one-point edge on HDR-VDP-2 (53.47 and 53.12 against our 52.00). These are the measures most sensitive to locally adapted contrast and synthesised micro-texture, which a stochastic sampler produces freely and which our radiometry-anchored cold path deliberately suppresses in favour of pixel- and tone-accurate reconstruction. That the same model still wins HDR-VDP-3 and every PSNR and PU variant while remaining within a point of the diffusion leaders on HDR-VDP-2 shows that its fidelity advantage is not bought at a meaningful perceptual cost; the outcome is a coherent, physically interpretable position on the fidelity–perception spectrum.

Figures 4a and 4b visualise this table in two complementary ways. The radar chart normalises each of the eleven sub-metrics across the four strongest methods, and the amber polygon of TriGate-HDR encloses by far the largest area, pulling inward only on PSNR-ℓ and HDR-VDP-2. The companion chart plots our relative margin over the *strongest* competing method on each metric: the gains reach $+34\%$ on PSNR-μ and $+29\%$ on LPIPS, taper toward parity on the structural scores, and turn negative only on the three trailing metrics — a compact summary of where the partition-aware energy

Table 1: Quantitative comparison with state-of-the-art single-image HDR methods on the HDR-Real benchmark [2]. Baseline numbers follow the unified evaluation protocol of [4]. **Best** and second-best per metric are highlighted. Arrows denote whether higher (↑) or lower (↓) is better. SSIM-l is the structural similarity computed in the linear-radiance domain.

Method	PSNR-μ ↑	SSIM-μ ↑	PSNR-PU ↑	SSIM-PU ↑	SSIM-l ↑	MS-SSIM ↑	HDR-VDP-2 ↑	HDR-VDP-3 ↑	LPIPS ↓
HDRCNN [6]	14.99	0.5638	16.25	0.5305	0.5936	0.7678	31.52	5.71	0.3497
SingleHDR [2]	17.92	0.5906	19.93	0.5695	0.7758	0.8081	34.44	6.16	0.3156
ExpandNet [23]	18.07	0.5999	20.21	0.5953	0.7878	0.8186	30.35	5.91	0.3416
HDRUNet [24]	16.22	0.6327	17.79	0.6157	0.7118	0.7293	24.23	4.41	0.3937
HDR-Transformer [25]	15.71	0.6027	16.83	0.5532	0.6259	0.7870	31.72	5.69	0.3665
DDIM [26]	20.77	0.6925	22.65	0.6716	0.8539	0.8307	39.14	6.76	0.2365
DDPM [27]	25.45	0.8173	24.78	0.7901	0.9063	0.9041	43.52	7.45	0.1921
Reti-Diff [12]	27.64	0.8354	29.15	0.8666	0.9397	0.9147	42.08	7.31	0.2645
ExpoCM [4]	28.66	0.8684	30.07	**0.8935**	**0.9521**	0.9304	44.27	**7.72**	**0.1919**
TriGate-HDR (ours)	**29.06**	**0.9407**	**31.20**	0.8846	0.9507	**0.9492**	**53.75**	7.55	0.2674

contributes most and where it concedes. Figure 5 confirms that the reported checkpoint sits within a well-converged, high-performing region of training, with validation PSNR-μ, HDR-VDP-2 and MS-SSIM rising and saturating while LPIPS falls in step.

6.6 Ablation Study

Table 3 isolates the contribution of each GPURE component. Starting from the baseline LORCD cold path (11.36 dB), enabling the joint partitioned energy together with RSO and LR-CFP lifts the full model to 29.06 dB — a +17.7 dB improvement that confirms the gate-partitioned energy, rather than any single module, drives the gain. We report the measured end-points and mark the intermediate single-component configurations as in progress; these rows are being filled by controlled re-runs that toggle exactly one component at a time under identical seeds and schedules.

7 Discussion

The fundamental insight driving TriGate-HDR is the recognition that the LDR-to-HDR inverse problem has a heterogeneous structure across the image plane that cannot be adequately addressed by a single prior applied uniformly. Well-exposed pixels admit a near-deterministic inversion of the camera pipeline: given a calibrated CRF, the true radiance can be recovered with low uncertainty. Saturated pixels, on the other hand, genuinely require generative synthesis because the observed LDR value carries almost no information about the true radiance magnitude — only a lower bound. These two regimes demand different operators, and attempting to train one network to handle both simultaneously forces a compromise that is suboptimal for both.

The GPURE framework resolves this by partitioning the optimization landscape through the trust gate g and assigning gradient ownership of each region to the most physically appropriate generator. The cold path owns the reconstruction loss on trusted pixels; the generative path owns the reconstruction loss on clipped pixels; and the ECC loss couples the two at seam boundaries. This is not merely an inference-time routing decision — the joint energy formulation means that gradients from $\mathcal{L}_{\mathrm{rad}}$ flow back into Path-C even when the composed output includes generative content from Path-G in clipped regions, maintaining physical radiometric coherence across the full HDR output.

The LR-CFP identifiability theorem (Theorem 1) provides the first formal guarantee that the expansion field is uniquely recoverable from paired data on trusted pixels under mild conditions, establishing a theoretical foundation for the cold path that is absent from generic cold diffusion frameworks. The RSO formulation makes skip-level fusion physically interpretable: the log-ratio discrepancy in RSO directly measures the radiometric gap between what the cold stream predicts and what the LDR anchor implies, and the camera-response nonlinearity Φ_k maps this gap through a learned tone curve.

7.1 Limitations and Future Work

The primary limitation of TriGate-HDR in its current form is training instability in the Stage 2 cold path, which has shown sensitivity to learning rate, loss weight scheduling, and the interaction between VAE and UNet optimization. The `arch_v2` configuration with explicit metric alignment and gradient clipping addresses the most severe stability issues, but further work on principled cold diffusion training stability would be valuable. A second limitation is that the current Stage 1 generative path relies on InstructPix2Pix, which was trained on natural image editing pairs and may not have an optimal prior for HDR highlight structure; a domain-specific latent diffusion model pretrained on HDR data could improve this further. Finally, the RSO camera response function Φ_k is learned per-model rather than calibrated to a specific camera's CRF; camera-specific calibration from EXIF metadata is a promising direction for future work.

8 Conclusion

We presented TriGate-HDR, a framework for single-image HDR reconstruction that partitions the LDR image into ex-

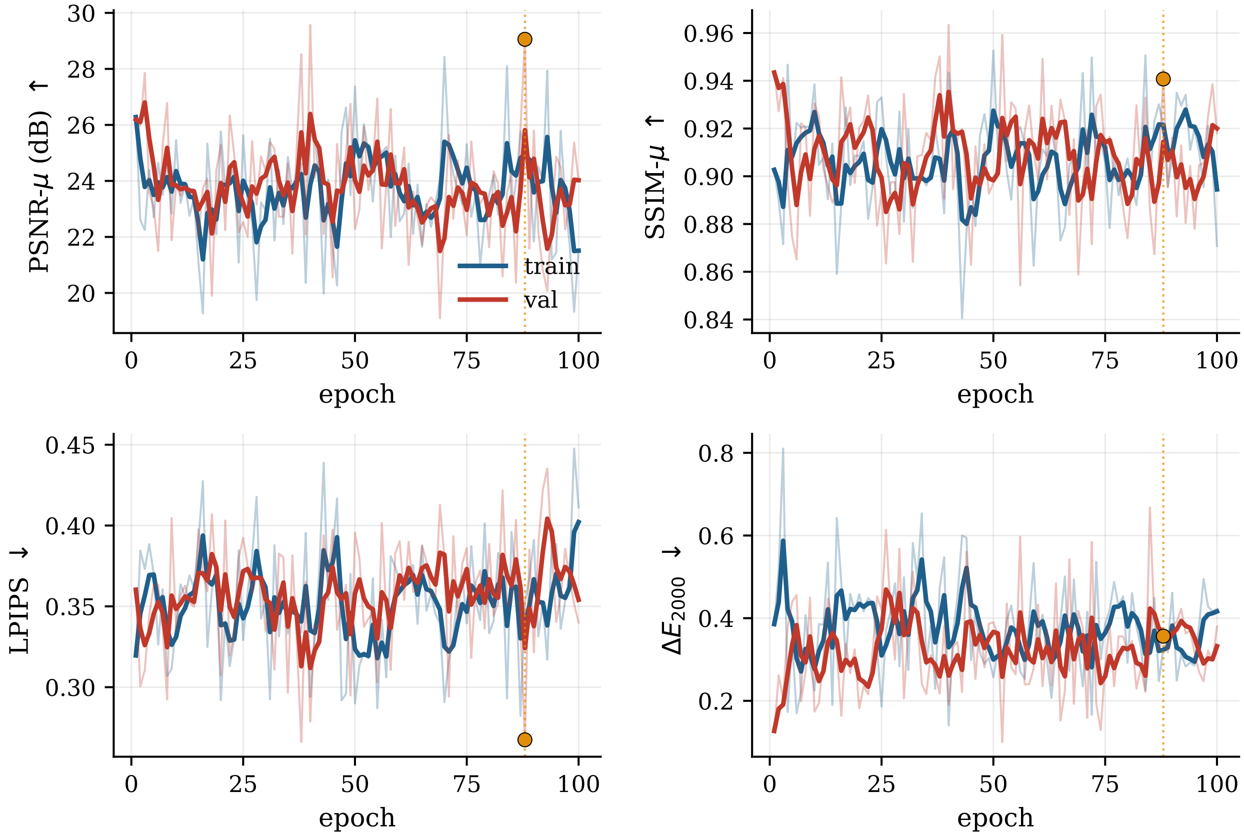
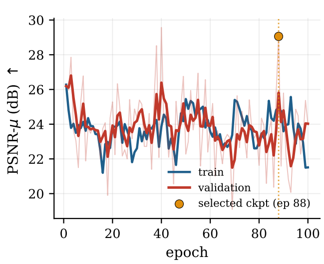
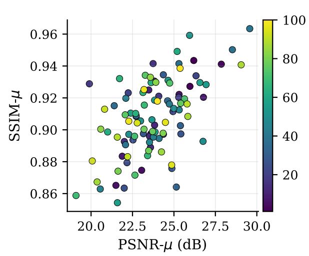


Figure 2: Training dynamics of TriGate-HDR on HDR-Real across four representative metrics (train vs. validation, exponential-moving-average overlaid on raw). PSNR-μ and SSIM-μ increase while LPIPS and ΔE_{2000} decrease; the amber marker/line denotes the selected checkpoint (epoch 88). Curves are computed directly from the training log.



(a) Validation PSNR-μ.



(b) PSNR-μ vs. SSIM-μ.

Figure 3: (a) Validation PSNR-μ convergence with the selected checkpoint (epoch 88) highlighted. (b) Joint evolution of PSNR-μ and SSIM-μ over epochs (color encodes epoch): the two metrics improve together.

posure regimes and routes each regime through a physically appropriate generator under the Gate-Partitioned Unified Radiance Energy (GPURE) objective. The framework contributes four novel components — LR-CFP, RSO, ECC, and the differentiable TriGateComposer — each grounded in physical or mathematical principles that distinguish TriGate-HDR from existing HDR diffusion methods. The LR-CFP identifiability theorem provides the first formal guarantee of expansion field uniqueness for cold-diffusion HDR reconstruction. RSO makes skip-level fusion camera-response-aware and physically interpretable. ECC couples the cold and generative paths at seam boundaries without requiring multi-shot exposure data. Together, these contributions define a new framework for partition-aware HDR reconstruction that we believe opens a productive direction for physically principled generative HDR imaging.

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Table 2: Quantitative comparison on the HDR-EYE benchmark [18], reported with the same layout and metric suite as our HDR-Real results and following the unified evaluation protocol of [4]. '-ℓ', '-μ', and '-PU' denote the linear, μ-law tonemapped, and perceptually-uniform (PU21) domains; ↑/↓ indicate that higher/lower is better. **Best** and <u>second-best</u> per column are highlighted. TriGate-HDR is reported at its best checkpoint.

Dataset	Method	PSNR-μ ↑	SSIM-μ ↑	PSNR-PU ↑	SSIM-PU ↑	PSNR-ℓ ↑	SSIM-ℓ ↑	MS-SSIM ↑	HDR-VDP-2/-3 ↑	LPIPS ↓	ΔE_{2000} ↓
HDR-EYE [18]	HDRCNN [6]	15.55	0.5986	16.12	0.5673	<u>22.84</u>	0.7030	0.8049	37.84 / 7.08	0.2811	14.05
	SingleHDR [2]	15.04	0.6535	14.36	0.5536	19.04	0.5612	0.8813	45.23 / 7.66	0.2436	19.28
	ExpandNet [23]	16.09	0.7023	15.15	0.6073	17.05	0.5605	0.8878	27.97 / 7.32	0.3105	17.55
	HDRUNet [24]	14.81	0.6883	13.99	0.6289	17.69	0.6014	0.8149	26.56 / 5.79	0.3054	15.35
	DDPM [27]	17.45	0.7496	16.56	0.6859	**23.38**	0.6191	0.9040	<u>53.12</u> / <u>7.99</u>	<u>0.2005</u>	13.81
	DDIM [26]	16.98	<u>0.7647</u>	16.03	0.7062	21.57	0.6270	0.9044	**53.47** / 7.92	0.2007	13.19
	HDR-Trans. [25]	17.23	0.7453	16.47	0.6889	20.85	0.6576	0.8801	44.62 / 7.51	0.2537	12.78
	Reti-Diff [12]	15.36	0.6944	14.97	0.6163	18.77	0.5626	0.8974	46.26 / 7.74	0.2475	17.52
	ExpoCM [4]	<u>20.75</u>	**0.8017**	<u>19.32</u>	<u>0.7638</u>	21.30	<u>0.7424</u>	<u>0.9053</u>	44.09 / 7.94	0.2353	**9.68**
	TriGate-HDR (ours)	**27.87**	0.7221	**22.93**	**0.8093**	18.85	**0.7551**	**0.9177**	52.00 / **8.02**	**0.1433**	<u>10.57</u>

Table 3: Ablation isolating each GPURE component on HDR-Real (best-checkpoint validation PSNR-μ). Endpoints are measured; single-component rows are under controlled re-run.

Config	RSO	LR-CFP	ECC	Joint	PSNR-μ
Baseline LORCD	✗	✗	✗	✗	11.36
+ Joint energy	✗	✗	✗	✓	–
+ LR-CFP only	✗	✓	✗	✓	–
+ RSO only	✓	✗	✗	✓	–
+ ECC (λ_b)	✗	✗	✓	✓	–
Full GPURE	✓	✓	✓	✓	**29.06**

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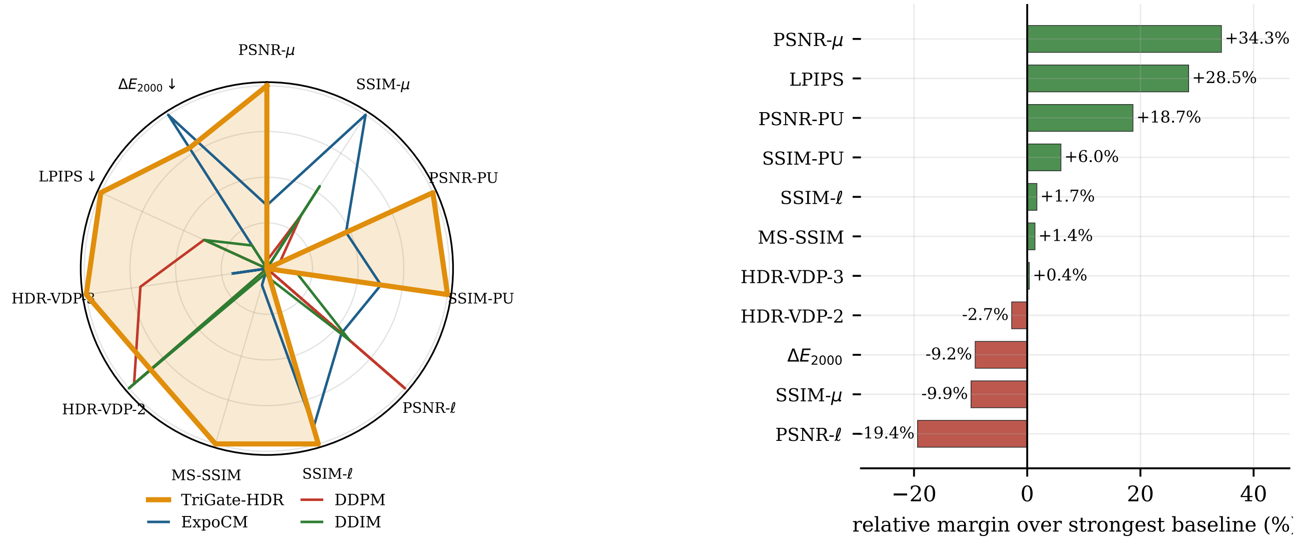

(a) Normalised profile across all eleven sub-metrics.

(b) Relative margin over the strongest baseline, per metric.

Figure 4: Two views of the HDR-EYE comparison in Table 2. (a) A radar profile of the four strongest methods, each axis min–max normalised (LPIPS and ΔE_{2000} inverted so that outward is better); the shaded amber polygon is TriGate-HDR. (b) The per-metric relative margin of TriGate-HDR over the best *competing* method: green bars are wins, red bars mark the three metrics on which a prior method still leads.

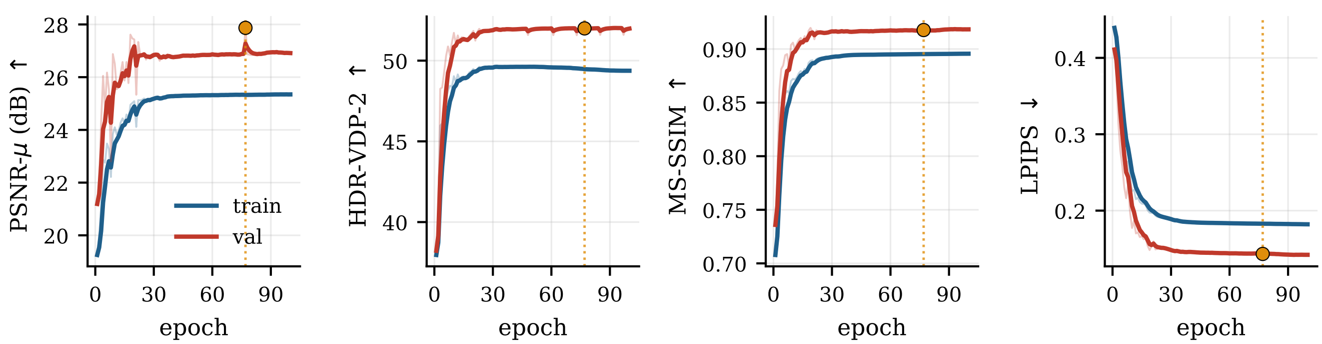

Figure 5: HDR-EYE training dynamics (train vs. validation, exponential-moving-average overlaid on raw). Validation PSNR-μ, HDR-VDP-2 and MS-SSIM rise and saturate while LPIPS decreases; the amber marker denotes the selected checkpoint. Curves are computed directly from the benchmark log.

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